Mitali Bhatia
D15C 16
MLDL

Experiment No 06

Aim: Application of K-Means and Hierarchical Clustering on sample datasets

1. DATASET SOURCE

Dataset Name: Mall Customers Dataset
Source: Kaggle
Link: https://www.kaggle.com/datasets/shwetabh123/mall-customers

This dataset is publicly available and commonly used for clustering and customer segmentation experiments.

---

2. DATASET DESCRIPTION

The Mall Customer dataset contains information about customers of a shopping mall.

Dataset Size:

- Total Records: 200 customers
- Total Features: 5 columns

Features:

1. CustomerID – Unique ID for each customer
2. Gender – Male/Female
3. Age – Age of customer
4. Annual Income (k$) – Annual income in thousand dollars
5. Spending Score (1-100) – Score assigned by mall based on spending behavior

Features Used in This Experiment:

- Annual Income (k$)
- Spending Score (1-100)

Target Variable:

There is **no predefined target variable** because clustering is an **unsupervised learning** technique.
The goal is to group similar customers into clusters based on income and spending behavior.

Dataset Characteristics:

- Numerical dataset (for selected features)
- No missing values
- Suitable for segmentation problems
- Small size dataset (good for visualization)

3. MATHEMATICAL FORMULATION OF THE ALGORITHMS

A) K-Means Clustering

K-Means divides data into K clusters by minimizing within-cluster variance.

1. Euclidean Distance

$$d(x, y) = \sqrt{\sum_{j=1}^{m} (x_j - y_j)^2}$$

Used to calculate distance between a data point and centroid.

2. Centroid Formula

$$\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

Where:

- μ_k = centroid of cluster k
- C_k = cluster k
- $|C_k|$ = number of points

3. Objective Function (WCSS)

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

Goal: Minimize total squared distance between data points and their respective centroids.

B) Hierarchical Clustering

Hierarchical clustering merges nearest clusters step by step.

1. Euclidean Distance

$$d(x, y) = \sqrt{\sum_{j=1}^{m} (x_j - y_j)^2}$$

2. Ward Method Formula

$$\Delta E = \frac{|C_1| \cdot |C_2|}{|C_1| + |C_2|} ||\mu_1 - \mu_2||^2$$

Ward method merges clusters that cause minimum increase in total variance.

4. ALGORITHM LIMITATIONS

K-Means Limitations

1. Value of K must be selected manually.
2. Sensitive to initial centroid selection.
3. Works only for numerical data.
4. Assumes spherical clusters.
5. Sensitive to outliers.

May not perform well when:

- Clusters have irregular shapes.
- Data contains extreme outliers.
- Dataset is highly non-linear.

Hierarchical Clustering Limitations

1. Computational complexity is high ($O(n^2)$).
2. Not suitable for very large datasets.
3. Once clusters merge, they cannot be split.
4. Sensitive to noise and outliers.

5. METHODOLOGY / WORKFLOW

Step 1: Import Libraries

Imported:

- NumPy
- Pandas
- Matplotlib
- sklearn.cluster
- scipy.cluster.hierarchy

Step 2: Load Dataset

```
data = pd.read_csv("Mall_Customers.csv")
```

Dataset loaded successfully.

Step 3: Feature Selection

Selected:

- Annual Income
- Spending Score

These features help in understanding customer purchasing behavior.

Step 4: Data Visualization

Plotted scatter plot of income vs spending score to observe distribution.

K-MEANS WORKFLOW

1. Calculated WCSS for K = 1 to 10.
2. Plotted Elbow graph.
3. Observed bend at K = 5.
4. Created KMeans model:

```
kmeans = KMeans(n_clusters=5, random_state=42)
```

5. Fitted model using `.fit(X)`
6. Extracted labels using `.labels_`
7. Extracted centroids using `.cluster_centers_`
8. Plotted clustered data with centroids.

HIERARCHICAL WORKFLOW

1. Generated linkage matrix:

```
linked = linkage(X, method='ward')
```

2. Plotted dendrogram.
3. Observed natural cut at 5 clusters.
4. Created AgglomerativeClustering model:

```
hc = AgglomerativeClustering(n_clusters=5)
```

5. Predicted cluster labels.
6. Plotted clustered data.

6. PERFORMANCE ANALYSIS

Since clustering is unsupervised, accuracy is not used.

K-Means Performance

- Elbow Method indicated optimal K = 5.
- WCSS decreased significantly up to K=5.
- Clusters clearly separated customers into:
    - High income – High spending
    - High income – Low spending
    - Low income – High spending
    - Low income – Low spending
    - Medium income – Medium spending

Clusters are compact and visually well-separated.

Hierarchical Clustering Performance

- Dendrogram clearly showed 5 major clusters.
- Ward method produced compact clusters.
- Results visually similar to K-Means.
- Good cluster separation observed.

---

7. HYPERPARAMETER TUNING

K-Means

Hyperparameters:

1. n_clusters
2. random_state
3. init
4. max_iter

Tuning Done:

- Tested K from 1 to 10 using Elbow Method.
- Selected K = 5 based on sharp bend.
- Used random_state = 42 for reproducibility.

Impact:
Proper selection of K improved cluster quality and separation.

Hierarchical Clustering

Hyperparameters:

1. n_clusters
2. linkage method

Tuning Done:

- Used Ward linkage.
- Selected 5 clusters by cutting dendrogram.

Impact:

Ward method minimized variance and produced compact clusters.

Code:

K-Means Clustering:

```
# ==============================

# K-MEANS

# ==============================

# Step 1: Import Required Libraries

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.cluster import KMeans

# Step 2: Load Dataset

data = pd.read_csv("Mall_Customers.csv")

# Step 3: Select Features

X = data[['Annual Income (k$)', 'Spending Score (1-100)']]

# Step 4: Visualize Original Data

plt.scatter(X['Annual Income (k$)'], X['Spending Score (1-100)'])

plt.xlabel("Annual Income (k$)")

plt.ylabel("Spending Score (1-100)")

plt.title("Original Customer Data")

plt.show()

# Step 5: Decide Number of Clusters using Elbow Method

wcss = []

for i in range(1, 11):
```

```
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)
plt.plot(range(1, 11), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
# Assume K = 5 (commonly chosen after elbow graph)
k = 5
# Step 6: Create KMeans Model
kmeans = KMeans(n_clusters=k, random_state=42)
# Step 7: Fit the Model
kmeans.fit(X)
# Step 8: Get Cluster Labels
labels = kmeans.labels_
# Step 9: Get Centroids
centroids = kmeans.cluster_centers_
# Step 10: Plot Clustered Data
plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=labels, cmap='rainbow')
plt.scatter(centroids[:, 0], centroids[:, 1],
            s=200, c='black', marker='X')
plt.xlabel("Annual Income (k$)")
```

```
plt.ylabel("Spending Score (1-100)")

plt.title("K-Means Clustering")

plt.show()
```

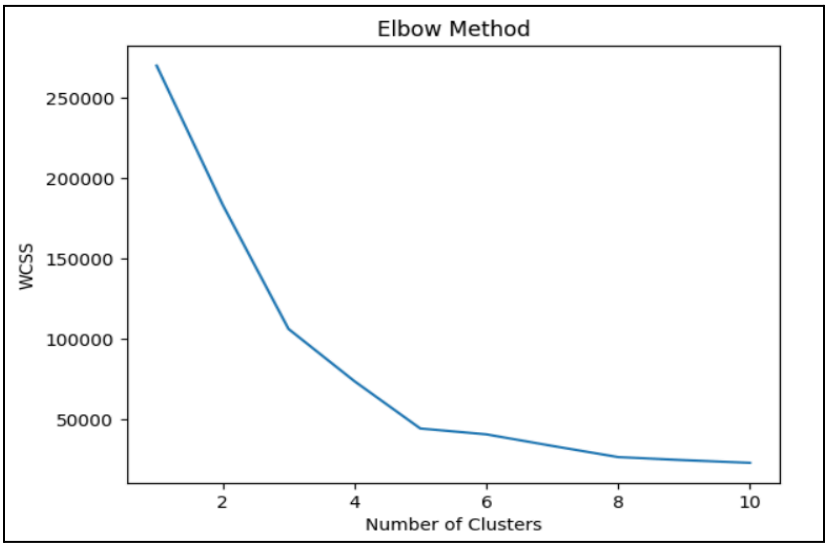
Hierarchical Clustering:

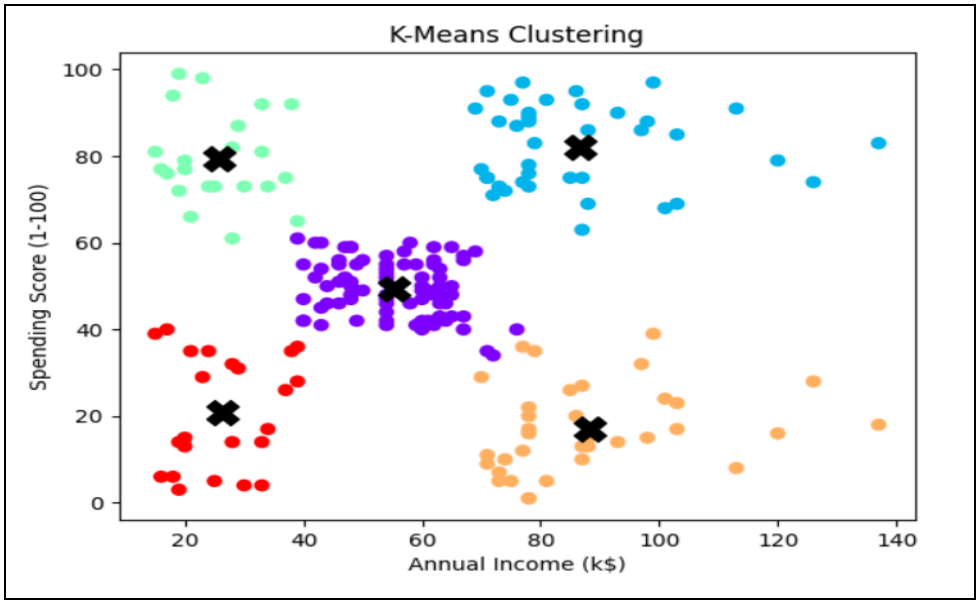
```
# ===================================

# HIERARCHICAL CLUSTERING

# ===================================

# Step 1: Import Required Libraries

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.cluster import AgglomerativeClustering

from scipy.cluster.hierarchy import linkage, dendrogram

# Step 2: Load Dataset

data = pd.read_csv("Mall_Customers.csv")

# Step 3: Select Same Features

X = data[['Annual Income (k$)', 'Spending Score (1-100)']]

# Step 4: Generate Linkage Matrix

linked = linkage(X, method='ward')

# Step 5: Plot Dendrogram

plt.figure(figsize=(10, 6))

dendrogram(linked)

plt.title("Dendrogram")
```

```
plt.xlabel("Customers")

plt.ylabel("Euclidean Distance")

plt.show()

# Assume number of clusters = 5 (after cutting dendrogram)

n_clusters = 5

# Step 6: Create Agglomerative Model

hc = AgglomerativeClustering(n_clusters=n_clusters)

# Step 7: Fit and Predict Labels

labels = hc.fit_predict(X)

# Step 8: Plot Clustered Data

plt.scatter(X.iloc[:, 0], X.iloc[:, 1],

            c=labels, cmap='rainbow')

plt.xlabel("Annual Income (k$)")

plt.ylabel("Spending Score (1-100)")

plt.title("Hierarchical Clustering")

plt.show()
```

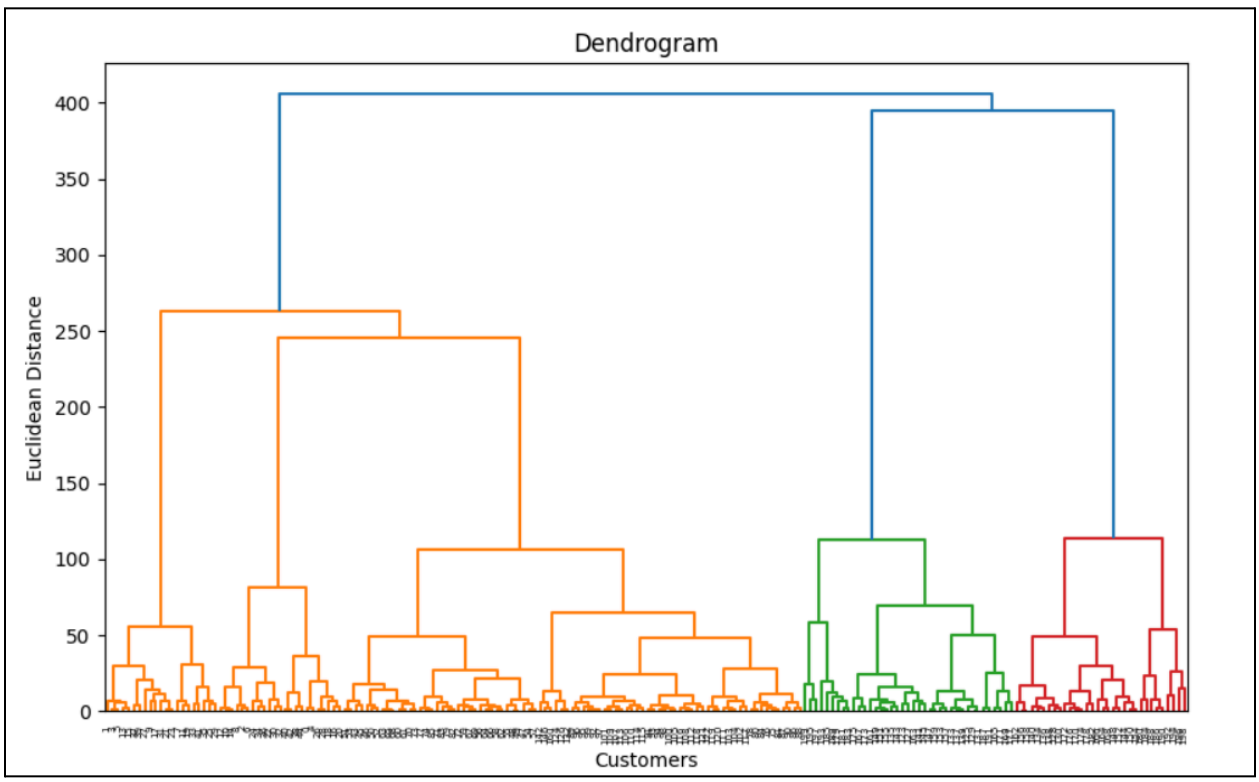
OUTPUT:

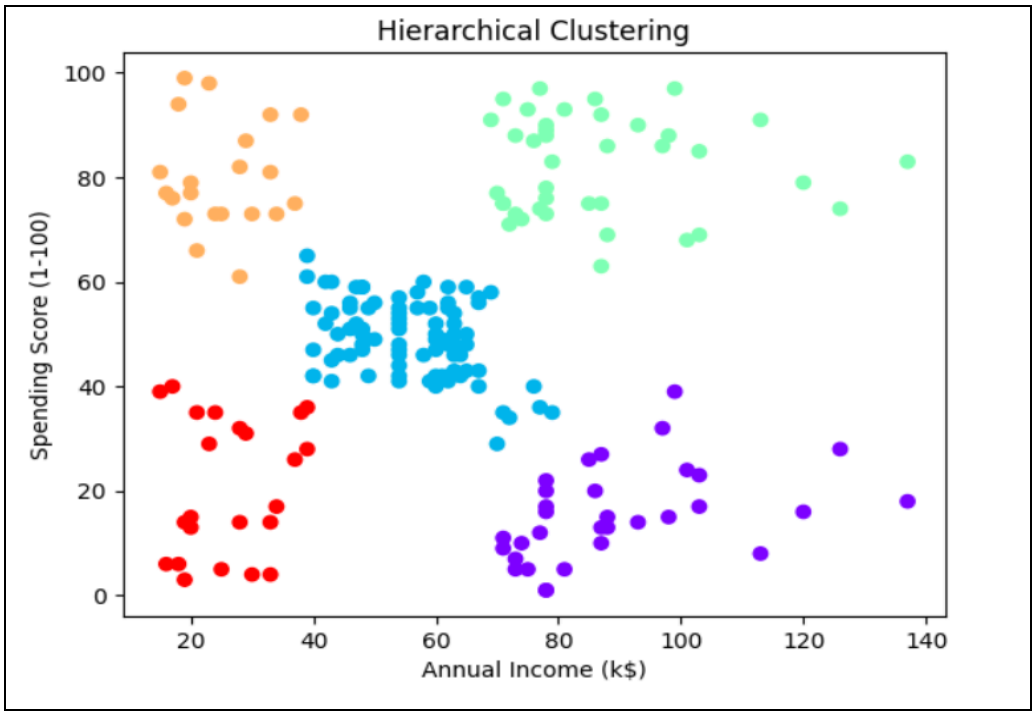
K-Means Clustering:





Hierarchical Clustering:





Conclusion:

In this experiment, K-Means and Hierarchical Clustering were applied to the Mall Customers dataset for customer segmentation. Using annual income and spending score, both algorithms successfully divided customers into five meaningful groups. K-Means was faster and easier to implement, while Hierarchical Clustering provided better visualization through dendrogram analysis. The Elbow Method and dendrogram helped in selecting the optimal number of clusters. Overall, clustering techniques proved useful in understanding customer behavior and can help businesses improve marketing strategies and decision making.